

SN65HVD230 CAN transceiver breakout

Ordered as ASIN **B0FLPTC5NM**: "Hoite SN65HVD-230 CAN Bus Module VP230 Communication Module Can Bus Transceiver Breakout Board", **pack of 5**.

These are the very common blue "VP230" clone boards. They carry a SN65HVD230-compatible 3.3 V CAN transceiver, a 120 ohm termination resistor, and a 6-pin 2.54 mm header.

Only one is needed for this project. The ESP32 is a single node on the bus; the other four are spares.

The IC

TI SN65HVD230, 3.3 V CAN transceiver, ISO 11898-2 physical layer, up to 1 Mbit/s. Datasheet:

<https://www.ti.com/product/SN65HVD230>

Pin	Name	Meaning
1	D	TXD - driver input, from the MCU
2	GND	Ground
3	Vcc	3.3 V only
4	R	RXD - receiver output, to the MCU
5	Vref	Vcc/2 reference output, normally left unconnected
6	CANL	Bus low
7	CANH	Bus high
8	Rs	Mode select - see below

Supply voltage

3.3 V. Do not feed this board 5 V. The SN65HVD230 is a 3.3 V part; this is exactly why it suits the ESP32 directly, with no level shifting on D/R.

(Beware: the visually identical "VP231"/"VP232" boards and the MCP2551 boards are 5 V parts. Check the silkscreen.)

Rs pin - slope control

Rs connection	Mode
Tied to GND (or via a resistor < 1.6 kohm)	High speed - what we want
Via 10-100 kohm to GND	Slope control, slower edges, less EMI
Tied to Vcc	Low-power standby - the driver is off

Most breakouts already tie Rs to GND through a resistor, but some expose it on a pad or jumper. **Verify before use:** if Rs floats or sits at Vcc, the transceiver will not transmit and the bus stays silent.

Termination

CAN is a terminated transmission line. The rule is **exactly two 120 ohm resistors, one at each physical end of the bus** - never more.

These breakouts almost always have a 120 ohm resistor fitted on-board. With three Daly BMS units (which may terminate internally too) plus this board, it is easy to end up with four or five terminators, which loads the bus down until nothing communicates.

Check and fix before connecting anything live:

1. With power off and the node disconnected from the bus, measure across CANH-CANL on each node. About 120 ohm means that node is terminated; open circuit means it is not.
2. Decide the two physical ends of the run. Leave those two terminated.
3. On every node in the middle, remove (or cut the jumper for) the 120 ohm resistor.
4. Re-measure across the assembled, unpowered bus: it should read about **60 ohm**. Around 40 ohm means three terminators; 120 ohm means only one.

Wiring to the ESP32

Breakout pin	Goes to
3V3 / VCC	Board 3V3 on the 2 x 8 header
GND	Board GND
CTX / D / TXD	ESP32 TWAI TX GPIO
CRX / R / RXD	ESP32 TWAI RX GPIO
CANH	Bus CANH (twisted pair)
CANL	Bus CANL (twisted pair)

TWAI TX/RX route through the ESP32 GPIO matrix, so any free GPIO works. See [wiring.md](#) for the pin choice and bring-up procedure.

Current draw is about 10-20 mA - comfortably within what the board's 3V3 rail can supply.